Adaptations

A Science A–Z Life Series
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Written by Ron Fridell

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KEY ELEMENTS USED IN THIS BOOK
The Big Idea: Plant and animal species must adapt in response to changes in the environment. These changes range from global to microscopic and may include changes in the climate, populations of other species sharing the same habitat, and the availability of essential resources for survival. Physical and behavioral adaptations are natural occurrences, not deliberate choices or momentary decisions made by individuals. Successful traits and behaviors allow organisms to survive and reproduce, and are passed on to offspring. These traits and behaviors become adaptations in future generations. Species often have many different adaptations, which has led to incredible diversity in nature.
Key words: adapt, adaptation, animals, behavior, birds, blowhole, canopy, cell, characteristics, climate, Darwin, desert, drip tip, environment, evolution, extinct, gene, generation, habitat, humans, inherited, instinct, mutation, naturalist, organism, physical, plants, predator, rainforest, reflex, scientists, species, survival of the fittest, survive **Key comprehension skill:** Main idea and details *Other suitable comprehension skills:* Classify information; identify facts; compare and contrast; elements of a genre **Key reading strategy:** Using a glossary and bold-faced words *Other suitable reading strategies:* Using a table of contents and headings; ask and answer questions; connect to prior knowledge; summarize; visualize

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Cactus spines are an adaptation to protect the plant from being eaten.

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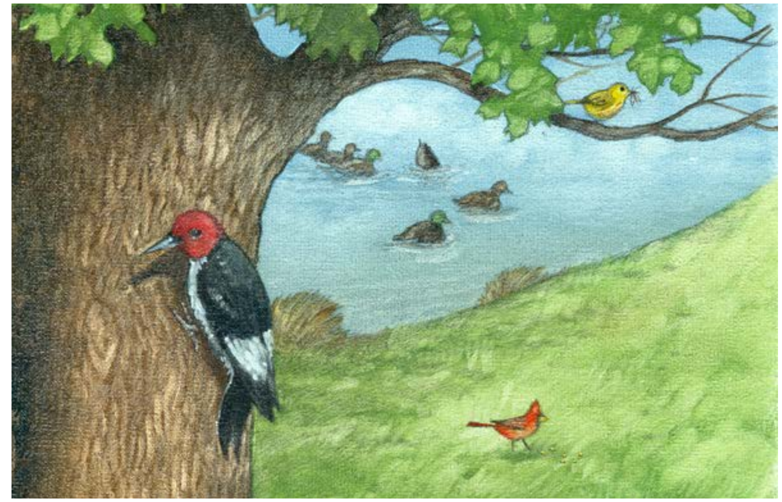


Introduction

Afternoon sunlight filters through the trees in a city park. A small bird called a *warbler* sits on a high tree branch, singing a beautiful song. Suddenly, he stops and dives to catch an insect out of midair. The insect struggles, but it can't escape. The warbler returns to his perch to enjoy his snack. Down on the ground, a bright red cardinal cracks open seeds with his hard bill. A quick, hard *rat-tat-tat-tat* sound splits the air, and splinters fly as a woodpecker drives her hard, chisel-like beak into a tree. On the hunt for grubs, she drills through the bark as a jackhammer breaks up concrete. On a pond nearby, some ducks float along lazily. Their heads dip down into the water and back up, their wide bills dripping with water.

These birds are all looking for a meal, but what they eat is different. Because their food is different, their beaks have different shapes. The warbler's beak—thin and pointed, like tweezers—is designed for grabbing insects. The cardinal's short, hard beak is made for cracking seeds. The woodpecker uses its long, strong beak to bore into wood, where insects lay their eggs. Ducks have wide bills for straining plants and small fish out of the water.

Over hundreds of thousands of years, these birds' beaks changed to help them gather the food they need to survive. All plant and animal **species** have **adaptations**—changes that help them survive in their **habitat**. Why do certain adaptations develop, and how do they help each plant or animal? Let's find out.

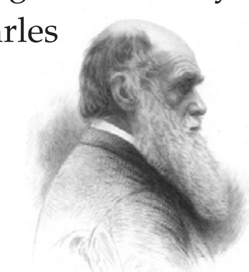
Each bird's beak is adapted to its habitat and food source.

Survival of the Fittest

Species **adapt** to changes in their environment in order to survive. How do adaptations develop? Individual **organisms** within a species are born with differences called **mutations**. These mutations may change the way an animal or plant grows and how it behaves.

Some individuals have beneficial traits and **behaviors** to help them live in a certain environment. These individuals are able to survive and reproduce. Then those traits and behaviors are passed on to the next **generation**. Individuals that do not have the beneficial traits and behaviors may not survive. Therefore, they may not reproduce. Over many generations, more and more organisms are born with the beneficial traits and behaviors. These changes become adaptations for the whole species.

This notion of species adapting to survive is known as natural selection, or **survival of the fittest**. The person who first brought this theory to the world's attention was Charles Darwin, a British **naturalist**. He wrote about it in his book *On the Origin of Species by Means of Natural Selection*, which was published in 1859.

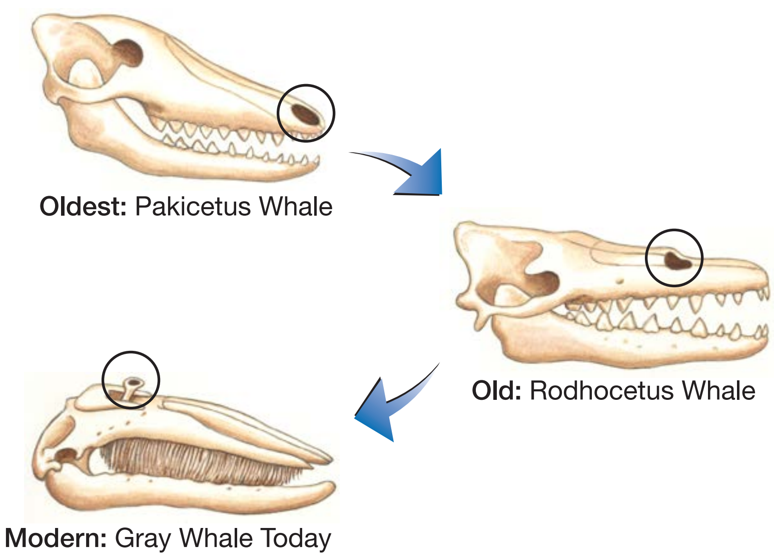


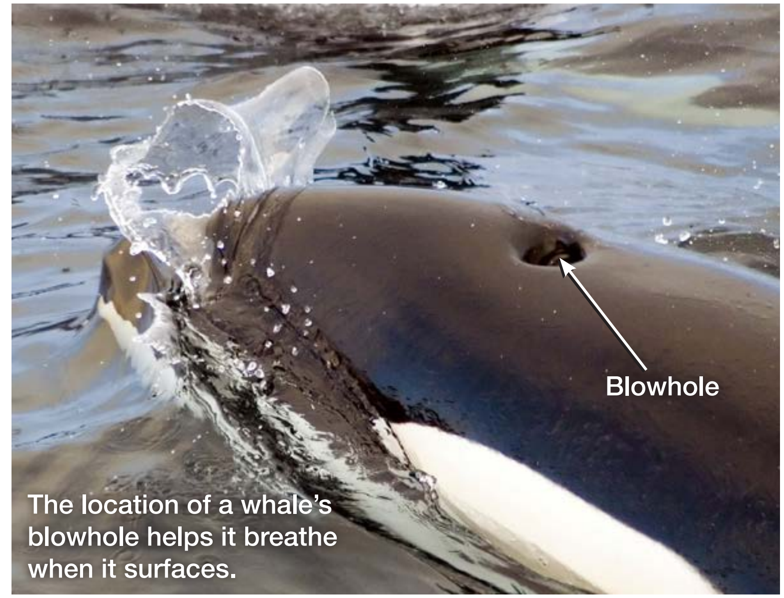
Charles Darwin

Let's consider an example of one adaptation and how it developed. Millions of years ago, all whales had teeth and breathed out of holes at the end of their snouts. Over many generations, some whales developed a breathing hole farther up their head. These whales could breathe more easily in deep water. They could rise to the water's surface and take a breath without sticking their snouts above the water.

As whales began to spend more time far out in the ocean, away from shore, more whales with higher breathing holes survived. Their babies also had breathing holes high on their heads. The high breathing hole became a **characteristic** of, or feature that identifies, a whale.

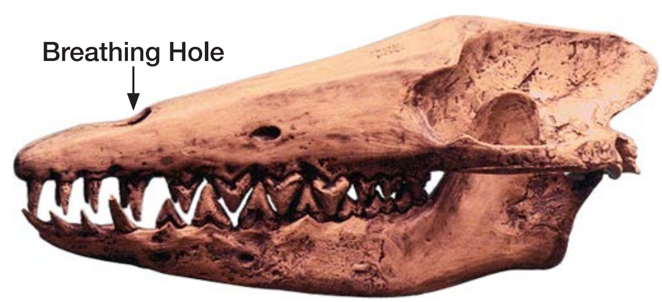
Evolution of the Whale's Nose





The location of a whale's blowhole helps it breathe when it surfaces.

Today, a whale's nose, called a **blowhole**, is atop its head. The blowhole makes it much easier for the whale to breathe. When a whale arches its body, its blowhole breaks the water's surface briefly to allow the whale to take a breath. Then, with its lungs filled with a fresh supply of air, it flexes its tail and returns to the ocean's depths.



This skull is from a very ancient whale. Its breathing hole was farther forward than a modern whale's.

What becomes of species that cannot adapt to their changing environments? They become **extinct**, meaning they no longer exist at all. For every species alive today, perhaps a thousand more became extinct before humans existed. We know of these extinct species only through fossil records.

An artist's drawing of various dinosaur species that are now extinct.

Dinosaurs became extinct 65 million years ago after living on Earth for about 165 million years. Why? Most scientists now agree that a giant meteor hit Earth. It created so much dust that it blocked sunlight, which caused the dinosaurs' food supply to die. Many dinosaur species became extinct because they could not adapt to Earth's quickly changing climate.

Desert plants are spaced widely apart.

Plant Adaptations

Plant species also have adaptations that allow them to survive and reproduce in their environment. For instance, they must get enough water and sunlight to make their own food.

Let's compare the adaptations of desert plants and rainforest plants. In deserts, water is a scarce and precious resource. Plants in deserts are spaced widely apart so they can share their environment's limited supply of water.

Rainforest plants, on the other hand, get more than enough water. Rainforests get about 200 centimeters (80 in.) of rain per year, compared to less than 25 centimeters (10 in.) in deserts. Heavy rainfall is why the leaves of some rainforest plants have *drip tips*—sharp points, which help the plants shed water quickly. Too much water can weigh down the leaves, block photosynthesis, and cause fungus or bacteria to grow.

In the desert, water is scarce, but there is plenty of sunlight—too much sunlight for some plants. That is why the desert *Haworthia* grows mostly underground, where things are cooler. However, it must collect some sunlight to make food, so the tips of its leaves have clear "windows," which peek out from the sand to admit light.

Drip tips help plants shed water quickly.

"Windows" in the *Haworthia*'s leaves let in light.

For many rainforest plants growing near the ground, the problem is too little sunlight. Tall trees form a thick canopy overhead to capture sunlight. This canopy leaves the forest floor in shade. Woody vines called *lianas* have adaptations that allow them to survive in these conditions. Some climb the tallest trees to get to the available light above the canopy. Other vines start life high up in the canopy and then send their roots earthward.

This vine is climbing a tree in the rainforest to reach sunlight.

Do You Know?

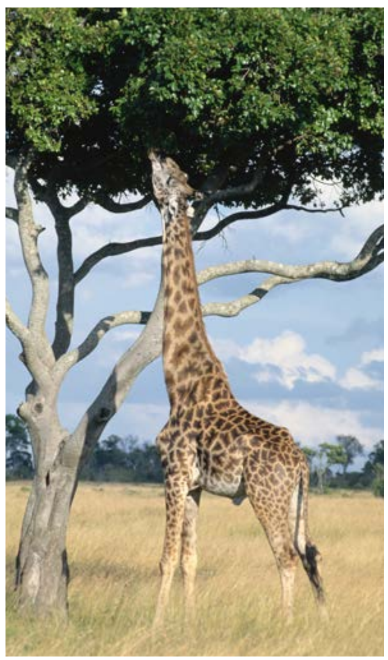
Plants cannot see or hear, but they can still sense changes in their environment. The keys are chemical substances within plant cells. These chemicals react to heat, light, moisture, temperature, and other factors that affect seed germination and plant growth. Depending on conditions, the chemicals signal the plant to grow taller, to flower, or perhaps to hold back growth and wait for a better time.

Animal Physical Adaptations

Plants make their own food, but animals must hunt for theirs. Over millions of years, species evolve and develop adaptations that help them find food and escape the animals that hunt them.

The giraffe's neck is a good example. Giraffes born with longer necks were able to reach higher into trees to gather more food than their shorter-necked relatives. They also had a better view of approaching predators. These long-necked advantages helped them survive and reproduce. Therefore, over time, more and more giraffes with longer and longer necks were born.

The giraffe's neck is an example of a physical, or structural, adaptation. *Physical adaptations* help animals adjust to their climate and landscape in all sorts of interesting ways.



Giraffes use their long necks to reach food.

These camels are traveling in a caravan across the desert sand.

Camels also have physical adaptations. These desert dwellers have an extra set of eyelids that are transparent. Deserts often have fierce sandstorms. Camels lower their transparent eyelids during these storms. The see-through lids protect camels' eyes from the stinging sands while still allowing them to journey through the desert. They can still find food and avoid predators, even through the blowing sand.

A camel eye has a clear eyelid.

Do You Know?

Before four-wheel-drive vehicles came along, camels were the chief means of transport in deserts. With their transparent eyelids and long legs, they are well adapted for desert travel. In long lines known as *caravans*, they can carry goods across desert sands where there are no roads or towns. This accounts for the camel's nickname: "ship of the desert."

Australia's koala bears are well suited to their environment. They spend much of their time in eucalyptus trees eating the leaves. A large gap separates their first and second fingers, and their big toe is set at a wide angle from each foot. These physical features help make koalas skilled tree climbers.

The giraffe's neck, the camel's eyelids, and the koala bear's fingers and toes are examples of physical adaptations. These are **inherited** characteristics, meaning they were passed down from parents to offspring.

Koalas have big gaps between their toes so they can grab branches.

Behavioral Adaptations

In addition to physical adaptations, animals have *behavioral adaptations.* Behavior describes how an animal acts and reacts to its environment. The simplest form of inherited behavior is a **reflex**, such as a frog jumping when it's touched. A reflex is a purely automatic reaction.

An **instinct** is a more complex inherited behavior. A sea turtle digging a hole in the sand to lay its eggs and a bird migrating south for the winter are both acting on instinct. These are behaviors that an animal just naturally knows it should do, without being taught.

In contrast, *learned behavior* changes as a result of experience. For example, you can train a dog to obey commands, and a goldfish can learn to swim to the surface when it sees a light. Chimpanzees can learn to use tools, and birds can teach their young new songs.

Some behaviors help animals attract a mate—a male peacock fanning its colorful feathers, for instance.


Male peacocks display their tails to attract females.

Or there's the deep-sea anglerfish's method. The males have large nostrils and a highly developed sense of smell, which they use to locate females, who cooperate by releasing a scented chemical for the males to follow.

Some behaviors help protect against predators. That's why many animals, such as flamingos and wildebeests, live and move about in large groups.

Other behaviors include cooperative hunting in lions, the digging behavior of rodents, and the ability of bees to produce honey.


A pufferfish inflates its body to look bigger to its predators.

Some behavioral adaptations puzzle scientists. They can't decide whether the behavior was inherited at birth or learned later.

One group of scientists set out to explore whether bird songs are inherited or learned behaviors. They studied the songs of birds raised in normal conditions—in a group with parents. These birds' songs sounded just like their parents' songs. The scientists compared their songs with the songs of birds that lived in isolation. These isolated birds grew up knowing how to sing, but their songs were simpler than the songs of their parents.


Birds sing simple songs by instinct, but learn complex songs from other birds.

The scientists concluded that singing is partly an inherited behavior because the isolated birds were born knowing how to sing. Since birds raised in a group adjust their songs to sound like their parents' songs, singing must also be partly learned.

Scientists have found that many behavioral adaptations are like the songs of birds. The adaptations are partly inherited and partly learned.

Human Adaptations

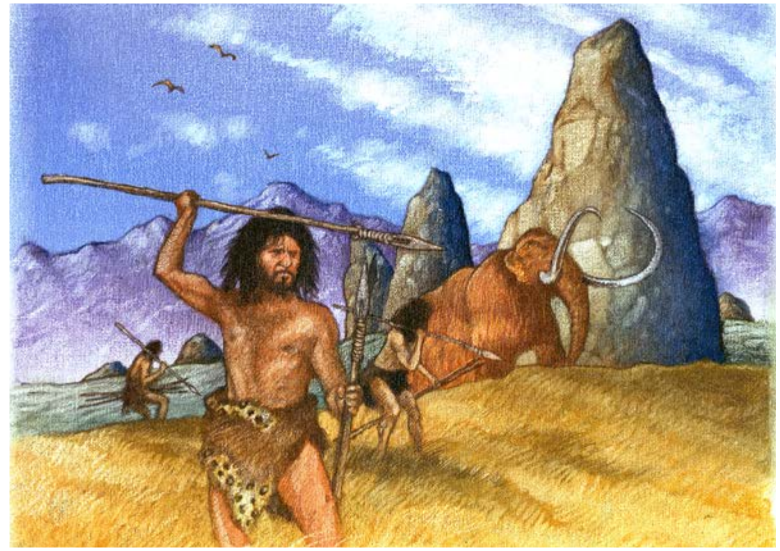
People have adaptations, too. For example, humans have developed the ability to sweat, which is a physical adaptation. When it is hot, the evaporating perspiration cools our skin. In cold climates, we shiver, which produces enough heat to warm us up for a short while. Sweating and shivering are both physical adaptations.

Sweating is an adaptation to help stay cool.

Shivering is an adaptation to help stay warm.

Do You Know?

Only mammals sweat. Primates sweat all over their bodies. Dogs and cats only sweat on their feet.



An artist's drawing of early humans using tools to hunt.

Humans also have behavioral adaptations that spring from our intelligence.

One of these adaptations is the ability to make tools. Early humans lived in a menacing environment. Without powerful jaws or sharp teeth and claws, they had to rely on their intelligence to survive. This led early humans to invent wood and stone tools for hunting, which gave them advantages over other animals.

WOWSER!

This crow bent a wire to turn it into a hook. That's the first evidence of birds making tools!

Many Words for the Same Thing

House – English

σπίτι – Greek

Haus – German

Casa – Spanish

Rumah – Indonesian

hus – Swedish

Bahay – Tagalog

房子 – Chinese

One of the most important human adaptations is our ability to use complex language to communicate. Other animals use sounds to communicate, but the songs of birds and the cries of monkeys are only simple signals. Humans have developed complex systems of sounds and symbols. Over 6,900 languages are spoken in the world today.

To survive, humans must satisfy the same basic needs as other animals. Other animals use a combination of mostly inherited physical and behavioral adaptations.

Humans, on the other hand, satisfy many of our needs through learned behaviors. To keep warm, we wear clothing, build fires, construct power plants, and live in insulated shelters. To get food, we grow fruits and vegetables and raise livestock. We learn to speak, read, and write. We use bikes, cars, and airplanes.

Adaptations Everywhere

Look at the photos on this page. What adaptations do you see? Which ones are physical adaptations and which are behavioral? Which of the behaviors are inherited and which are learned? How do these adaptations help each species?

Glossary

adapt	to change physical features or behaviors of a species in response to changes in the environment (p. 6)
adaptations	specific changes to physical features or behaviors of a species that help the species survive (p. 5)
behaviors	ways of acting in a certain set of circumstances (p. 6)
blowhole	a hole at the top of a whale or dolphin's head, through which it breathes (p. 8)
characteristic	a physical feature that helps identify an organism (p. 7)
extinct	no longer in existence (p. 9)
generation	all the organisms of a species born around the same time; the time between when a group of organisms and their offspring are born (p. 6)
habitat	the natural conditions and environment in which a plant or animal lives (p. 5)

inherited	passed on from parent to child (p. 15)
instinct	inherited behavior that leads an animal to act certain ways in certain situations (p. 16)
mutations	changes that living things are born with that may alter how they grow and what they can do (p. 6)
naturalist	a person who studies plants and animals (p. 6)
organisms	living things (p. 6)
reflex	a purely automatic response (p. 16)
species	a group of related organisms with characteristics that distinguish them from other groups of organisms (p. 5)
survival of the fittest	a theory explaining that the organisms best suited to live in a particular environment are those most likely to survive (p. 6)